

Moses Mulwa Francis

Full-Stack Python Developer · Django · REST APIs · VPS Infrastructure · AI Integration

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easyhostels.livify.co.ke (live product)

PROFILE

Full-Stack Python Developer with 2+ years building, shipping, and operating production web applications. Currently running EasyHostels — a live student housing marketplace with 150+ active users and 10+ registered landlords — on a self-managed VPS with Cloudflare and custom domain infrastructure. Experienced across the full delivery lifecycle: database design, secure API development with Django REST Framework, containerisation with Docker, and production DevOps on Linux servers. Mozilla Responsible Computing Challenge runner-up (2nd of 30 teams). Known for intellectual honesty, clean code, and a bias for shipping working software over perfect plans.

TECHNICAL SKILLS

Languages	Python · JavaScript (ES6+) · SQL · Bash / Shell
Backend	Django · Django REST Framework (DRF) · Flask · FastAPI (familiar) · Celery
Databases	MySQL · PostgreSQL · Django ORM · N+1 optimisation · schema design · migrations
Auth & Security	JWT · Google OAuth 2.0 (django-allauth) · RBAC · email token verification · env secrets management
APIs & Integr.	RESTful API design · DRF Serializers & ViewSets · Gmail API · Groq API · Google Sheets API · Weather APIs
DevOps & Infra	VPS admin (Truhost / Ubuntu) · Nginx · Cloudflare (DNS · SSL · proxy) · Docker · Docker Compose · systemd · SSH
Cloud & Hosting	cPanel · PythonAnywhere · Render · domain & MX record management · SSL certificates
Testing & CI	pytest · Git · GitHub · Pull Requests · CI/CD pipelines
AI / ML	Groq API · LLaMA 3.3 · OCR (pytesseract · pypdf) · Prompt Engineering · AI Agents (actively learning)
Frontend	HTML5 · CSS3 · Tailwind CSS · JavaScript Fetch API · DOM manipulation · Async/Await · Figma

PROFESSIONAL EXPERIENCE

Freelance Python Developer · InvoiceIQ — AI Invoice Processing Platform · Remote Feb 2026 – Mar 2026

- ▶ Built a full-stack invoice automation platform using Django and DRF, integrating the Groq API (LLaMA 3.3) to parse unstructured invoice data into validated, structured JSON — eliminating manual data entry in financial workflows.
- ▶ Engineered an OCR pipeline (pytesseract + pypdf) capable of extracting text from both scanned and digital PDF documents accurately.
- ▶ Designed a Human-in-the-Loop (HITL) review interface ensuring 100% data accuracy before records were committed to the database — critical for financial compliance.
- ▶ Implemented JWT authentication, Celery for asynchronous background processing, and structured audit logging; maintained API response times under 300ms for large document batches.

Stack: [Python](#) · [Django](#) · [DRF](#) · [MySQL](#) · [Celery](#) · [Groq API](#) · [LLaMA 3.3](#) · [OCR](#) · [JWT](#) · [Docker](#)

Junior Developer Intern · [Timnjiz Tech Company](#) · Kenya

May 2025 – Sep 2025

- ▶ Led backend architecture for a School Management System serving high-traffic student records; applied `select_related` and `prefetch_related` throughout to eliminate N+1 queries and reduce server overhead significantly.
- ▶ Collaborated directly with non-technical school staff to iterate on the frontend interface, translating real-world feedback into UX improvements that improved system adoption.
- ▶ Maintained 100% office infrastructure uptime through proactive hardware troubleshooting and PC maintenance.

Stack: [Django](#) · [MySQL](#) · [Python](#) · [Git](#) · [HTML/CSS](#)

PROJECTS

EasyHostels — Student Housing Marketplace ▶ [LIVE](#) · [easyhostels.livify.co.ke](#)

- ▶ Built and launched a production housing platform now operating at genuine scale: 10+ registered landlords and 150+ students on an active waiting list — organic demand with zero marketing spend.
- ▶ Identified a critical bottleneck in production (waitlist outpacing supply) and is currently architecting a real-time notification pipeline using Django signals and Celery to alert students the moment a matching property is listed — reducing drop-off and improving match velocity.
- ▶ Self-managed the complete production infrastructure: provisioned and hardened a Truhost Ubuntu VPS, configured Nginx as a reverse proxy, handled Cloudflare DNS routing and SSL termination, and managed the custom domain — zero reliance on managed hosting platforms.
- ▶ Designed a normalised relational schema (MySQL) for property listings, user inquiries, and automated matching; enforced referential integrity at both database and ORM level to prevent ghost listings and duplicate bookings.
- ▶ Implemented end-to-end authentication: Google OAuth (django-allauth), email token verification, and secure password reset — handling real user accounts and sensitive housing data in production.

Stack: [Django](#) · [MySQL](#) · [JavaScript](#) (Fetch API) · [Tailwind CSS](#) · [VPS](#) (Truhost) · [Nginx](#) · [Cloudflare](#) · [Ubuntu Linux](#) · [Celery](#) (in progress)

Decision Support System 🏆 [Mozilla RCC Runner-Up — 2nd of 30 teams](#)

- ▶ Built an AI-driven agricultural platform integrating multiple real-time weather and climate APIs to provide farmers with data-backed crop selection recommendations.
- ▶ Ranked 2nd out of 30 competing teams in the Mozilla Responsible Computing Challenge — recognised for innovation and responsible, real-world application of AI.

Stack: [Flask](#) · [Python](#) · [MySQL](#) · [JavaScript](#) · [REST APIs](#)

University of Embu CU — Evaluation & Feedback System

- ▶ Co-built (2-developer team) a centralised evaluation platform digitising the semester feedback loop for the University of Embu Christian Union's 1,000+ member base.
- ▶ Managed the full deployment lifecycle on cPanel: database configuration, SSL certificate management, DNS routing — achieving 24/7 student accessibility.
- ▶ Engineered a secure backend for structured data collection, enabling leadership to generate semesterly performance reports.

Stack: [Django](#) · [Python](#) · [MySQL](#) · [cPanel](#) · [HTML5](#) · [CSS3](#) · [JavaScript](#)

Christian Union Management System ▶ [In Development](#)

- ▶ Building an enterprise-grade centralised platform with RBAC, Google OAuth, and structured DRF endpoints to digitise organisational leadership and data management for 1,000+ members.
- ▶ Containerising the application with Docker and Docker Compose for consistent deployment and future scalability.

Stack: [Django](#) · [Python](#) · [DRF](#) · [Docker](#) · [Google Cloud API](#)

Bible Study Finder ▶ [Live](#)

- ▶ Built a lightweight serverless tool using JavaScript to fetch and display live data from Google Sheets, enabling real-time updates by non-technical administrators.

Stack: [HTML](#) · [CSS](#) · [JavaScript](#) · [Google Sheets API](#)

CERTIFICATIONS & TRAINING

AI Essentials & Generative AI for Software Developers · *Moringa School*

2025

- ▶ Intensive training on LLM integration, prompt engineering, and the ethical application of AI in software development workflows.

Python & Django Technical Workshops · *Various — Online & University-led*

Ongoing

- ▶ Advanced web architecture, secure API design, RESTful best practices, and production-ready Python code standards.

Leadership Development Certificate · *University of Embu Christian Union*

2024

- ▶ Formal training in team management, strategic planning, and ethical leadership — recognised by the university's student leadership body.

EDUCATION

BSc Information Technology · *University of Embu*

Completed 2026 · Awaiting Graduation

- ▶ All coursework complete; available full-time immediately.
- ▶ Key focus areas: Software Engineering · Database Management Systems · Data Communications & Networks · Big Data Analytics · Artificial Intelligence.
- ▶ Consistently applied academic concepts to production-grade systems — live housing platform, AI invoice system, award-winning decision-support tool.

PROFESSIONAL ENGAGEMENT & LEADERSHIP

Stakeholder — Engineering Curriculum Development Workshop · *University of Embu*

March 2026

- ▶ Invited as a key stakeholder to review and validate the BSc in Electrical, Electronic, and Automation Engineering curriculum alongside academic leaders and industry professionals.
- ▶ Provided student-centred insights on integrating modern technologies (Industrial Automation, IoT, Digital Systems) and advised on hands-on learning structures for graduate readiness.

Technical & Special Activities Coordinator · *University of Embu Christian Union*

2024 – Present

- ▶ Lead a 6-member technical subcommittee managing AV setup, digital system integration, and live technical operations for large-scale university events.
- ▶ Primary technical liaison between university administration, local authorities, and external stakeholders for event logistics.
- ▶ Collaborated with FOCUS Kenya to spearhead integrity and transparency campaigns, mentoring students on accountability and professional ethics.

Integrity & Environmental Advocate · *Community Outreach — Volunteer*

Ongoing

- ▶ Participated in regional outreach (Kangaru and Embu) campaigning for environmental conservation and ethical leadership within the resident community.

ADDITIONAL

- ▶ Languages: English (proficient) · Kiswahili (proficient)
- ▶ Interests: Technology for social good · Ethical AI application · Blockchain & IoT (continuous learning) · Piano performance & sound engineering
- ▶ Verification documents (certificates, leadership): Available on request and via supporting document links in full application.

REFEREES

- ▶ Timothy Njiru — CEO, Timnjiz Tech Ltd · Professional Supervisor (Internship) · +254 702 229 426

- ▶ Ben Birgen — Regional & Support Ministry Director, FOCUS Kenya · Leadership & Character Mentor · +254 708 610 505
- ▶ Betty Mutunga — Campus Ministry Facilitator, FOCUS Kenya (Embu & Chuka) · Technical Career Mentor & Leadership Advisor · +254 742 095 405